

CS3.301: Operating Systems and Networks

Lecture 5: CPU Scheduling

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Need for high level OS policies (Scheduling) – A parallel

- Multiple packaging workstations in a factory
- Limited robots carry materials from storage units to packaging workstations
- How will you schedule the picking of materials from storage units to workstations by robots such that some properties are optimized?
- Maximize throughput, minimize the maximum time required to process all the products
- Same is the situation within a computing system!



What we need to know to ensure good policy?

- How many products are there?
- What are the categories of each product?
- How much time does it take for each product to be processed?
- How frequently are the product orders coming in?

What does it mean?

- For scheduling, we need an idea of the **workload**
 - Assumptions of processes running in the system
 - Number of processes
 - RAM required
 - CPU utilization
 - Any Input/Output, if yes what kind?

Some workload assumptions

- Each process that is ready/needs to be executed and those executing called **Job!**
- Assumptions
 - Each job runs for **same amount of time**
 - All jobs arrive at the same time
 - All jobs **only use the CPU** (No I/O)
 - **Run time** or execution time of each job is **known**

How good is the policy?

- Metrics are used to measure the goodness of a policy
- **Performance metric:**
 - Turnaround time – Time difference between job completion time and the arrival time
$$T_{turnaround} = T_{completion} - T_{arrival}$$
 - Another metric is fairness – How fair is the scheduling?
 - May not go hand in hand with performance

Scenario 1: All assumptions intact

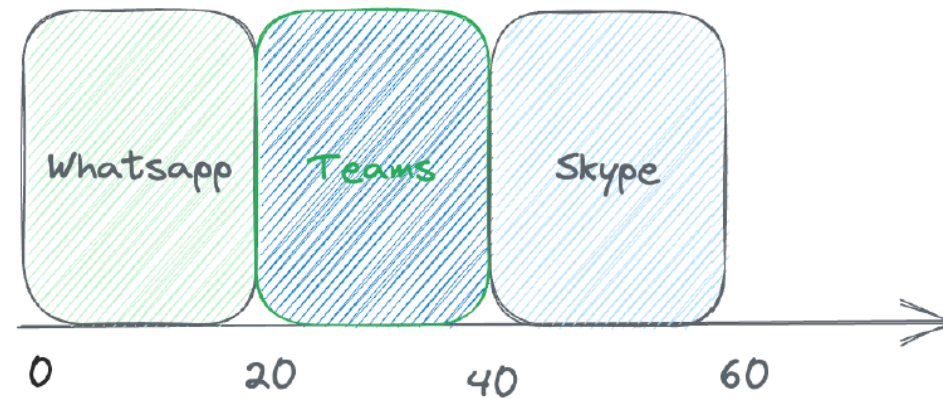
- Imagine 3 jobs – Whatsapp, Skype and Teams update arriving at the same time
- Each of them taking same time to complete

Process	Arrival	Time to Complete
Whatsapp (w)	~0	20
Skype (S)	~0	20
Teams (T)	~0	20

- How to go about this?

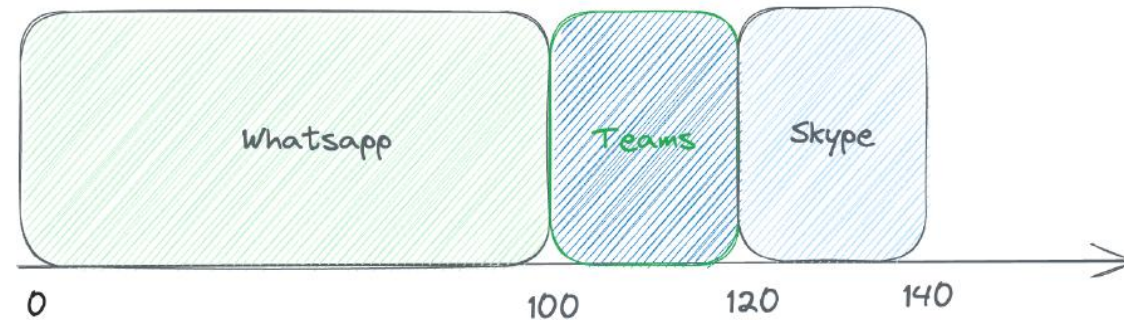
First Come First Serve (FCFS) Policy

- Most basic algorithm a scheduler can implement
- First one gets scheduled
- Assume that they arrive at the same time. Example $t = 0$



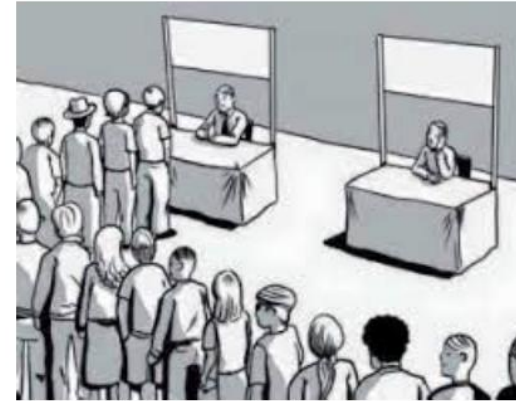
- Policy can pick a process randomly
- Average turnaround time = $(20 + 40 + 60)/3 = 40$

What if each job no longer runs for the same time?



- Average turnaround time = $(100 + 120 + 140)/3 = 120$

FCFS is not that great



- Waiting time can go very high
 - Convoy effect
 - Think about waiting in line to buy just one item

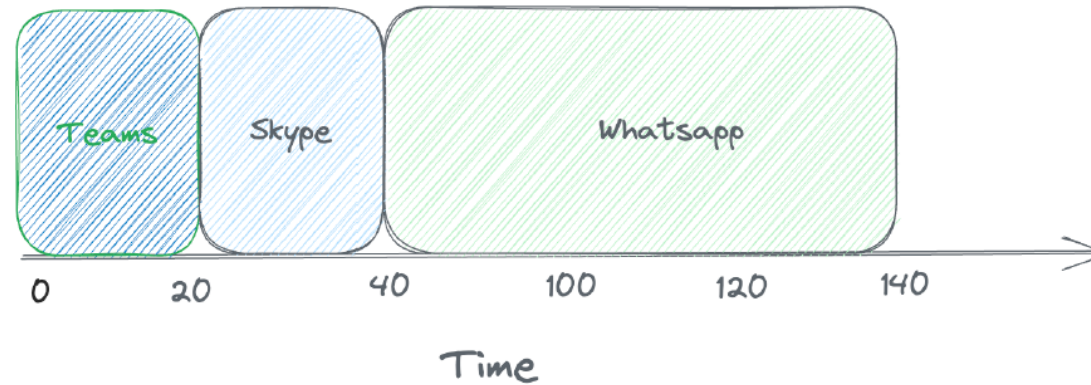
What if each job said what time it requires!

Shortest Job First (SJF) Policy

- Idea originating from Operations research
- Policy: Run the shortest job first

Process	Arrival	Time to Complete
W	0	100
S	0	20
T	0	20

SJF Policy



Average turnaround time = $(20+40+140)/3 = 66.3$

Thank You!!